

Reviewer Pack — Minimal Rank of Symplectic Geometry over XOR-AND Tree Width

Entry 28c209382021 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Symplectic Geometry over XOR-AND Tree Width

Entry ID: 28c209382021 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 19:49:37 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Complexity Theory: XOR-AND Tree Width

Statement:

[‘For any symmetric $n \times n$ matrix M representing a binary relation on $\{1, \dots, n\}$ that is both an XOR and AND gate, the minimal rank of the symplectic form induced by M is $\Theta(n^{\{1/2\}})$.’, ‘Equivalently, for any instance of the XOR-AND tree width problem with n nodes, the minimum dimension of a vector space in which the relation can be represented by a matrix of rank at most k , satisfies $\dim(\text{VectorSpace}) = \Theta(\sqrt{n/k})$.’]

Rationale (proposer’s reasoning):

[“Symplectic geometry is known for its applications in areas such as classical mechanics and algebraic geometry. The conjecture links it to XOR-AND tree width by suggesting a connection between the symplectic form’s rank and the complexity of representing a relation that can be modeled by XOR and AND gates.”, “This bridge may reveal new structures in both fields, potentially leading to more efficient algorithms for solving problems related to XOR-AND tree width.”]

Taxonomy category: MONOTONE_CLIQUE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f58965df2fcf5f6c

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a symmetric matrix M or XOR-AND tree width instance with n nodes and rank $k \leq n/2$, if the ratio of the symplectic form’s rank to $n^{\{1/2\}}$ is within $\pm 10\%$ of 1 for at least 90% of seeds, the conjecture is supported. If any seed produces a ratio outside this range or if less than 90% of seeds are within this range, the conjecture is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - symplectic geometry AND XOR-AND tree width - minimal rank symplectic form AND complexity theory XOR-AND tree width - vector space representation AND XOR-AND tree width AND symplectic geometry

Top relevant hits considered: - [<http://arxiv.org/abs/math/0601540v1>] Symplectic forms and surfaces of negative square - [<http://arxiv.org/abs/1811.09984v4>] Translated points for contactomorphisms of prequantization spaces over monotone symplectic toric manifolds - [<http://arxiv.org/abs/1007.4036v2>] Symplectic reduction of quasi-morphisms and quasi-states - [<http://arxiv.org/abs/1002.3099v3>] Tamed Symplectic forms and SKT metrics - [<http://arxiv.org/abs/0804.4433v1>] SPACE: the SPectroscopic All-sky Cosmic Explorer - [<http://arxiv.org/abs/1904.01500v1>] Neural Vector Conceptualization for Word Vector Space Interpretation - [<http://arxiv.org/abs/1910.01506v1>] Whistler instability stimulated by the suprathermal electrons present in space plasmas

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_matrix(n):
        M = [[random.randint(0, 1) for _ in range(n)] for _ in range(n)]
        for i in range(n):
            M[i][i] = 1
        return M

    def symplectic_form(M):
        n = len(M)
        I = [[1 if i == j else 0 for j in range(n)] for i in range(n)]
        S = [[M[i][j] ^ M[j][i] for j in range(n)] for i in range(n)]
        return S

    def matrix_rank(M):
        n = len(M)
        rank = 0
```

```

    for i in range(n):
        if any(M[j][i] != 0 for j in range(rank, n)):
            M[rank], M[i] = M[i], M[rank]
            for j in range(rank + 1, n):
                factor = M[j][i] / M[rank][i]
                for k in range(n):
                    M[j][k] -= factor * M[rank][k]
            rank += 1
    return rank

def xor_and_tree_width(n):
    # Placeholder implementation. This is a stub and should be replaced with actual logic.
    return n

n = random.choice([5, 10, 15, 20, 30, 40])
M = generate_matrix(n)
S = symplectic_form(M)
rank_S = matrix_rank(S)

expected_ratio = math.sqrt(n / rank_S)
observed_ratio = rank_S / math.sqrt(n)

metric_value = observed_ratio
instances_tested = 1
conjecture_holds = abs(observed_ratio - expected_ratio) <= 0.1 * expected_ratio
counterexample = "" if conjecture_holds else f"Observed ratio {observed_ratio}, expected ratio {expected_ratio}"

return {
    "metric_name": "Symplectic Rank Ratio",
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results) and support_fraction >= 0.9:
        first_failing_seed = next((seed for seed, result in enumerate(results) if not result["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"Observed ratio outside ±10% of expected\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient data or low support fraction")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_39406a6e.py", line 76, in <module>
    trial_result = run_trial(seed)
                   ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_39406a6e.py", line 53, in run_trial
    rank_S = matrix_rank(S)
             ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_39406a6e.py", line 40, in matrix_rank
    factor = M[j][i] / M[rank][i]
              ~~~~~^~~~~~
ZeroDivisionError: division by zero
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data, which means we cannot verify the conjecture's support or falsification based on the pre-registered criteria. | next: Re-run the test without errors to collect data for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12365
2	propose	ollama_remote	glm4:latest	0	0	14607
3	preregistration	ollama_remote	glm4:latest	0	0	6152
4	novelty	ollama_remote	glm4:latest	0	0	4577
5	novelty	ollama_remote	glm4:latest	0	0	6535
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17499
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9466
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9960
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9629
10	judge	ollama_remote	glm4:latest	0	0	11093

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 101883 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/28c209382021.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/28c209382021.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/28c209382021.tar.gz` (if generated)